

1. This Manual provides a basic guidance on doing econometric analysis using STATA.
2. *There are **NO** STATA-related questions in the quiz and final exam.
3. Doing linear regression analysis with one regressor:

- (a) Download the dta file (for example, “caschool.dta”) from NTU Learn.
- (b) Open the dta file (for example, “caschool.dta”) using the button “Open”.

Then, the following phrase is shown in the “Results” window:

use “/...../caschool.dta”, clear

This means that the dta file has been successfully imported into STATA.

- (c) You may browse the dta file using the button “Data Editor (Browse)”. The data set cannot be modified inside the “Data Editor (Browse)” [To modify the data by hand, we may use the button “Data Editor (Edit)”]; We need to be careful when using “Data Editor (Edit)”].

- (d) For the OLS regression with “caschool.dta”, type in the “Command” window (see Figure 1):

reg testscr str, robust

Then, the following table is shown in the “Results” window (see Figure 2):

Linear regression	Number of obs =	420
	F(1, 418) =	19.26
	Prob > F =	0.0000
	R-squared =	0.0512
	Root MSE =	18.581

testscr	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
str	-2.279808	.5194892	-4.39	0.000	-3.300945	-1.258671
_cons	698.933	10.36436	67.44	0.000	678.5602	719.3057

- (e) Note that the estimated slope and intercept, R^2 , (heteroskedasticity-robust) standard error, t^{act} , p -value, and confidence interval are reported in the table.

- (f) After running the OLS regression, we may ask STATA to generate the OLS predicted values by using the following command:

predict testscr_hat

Then, a new column called “testscr_hat”, which includes the OLS predicted value for each observation, is generated in the data set; This can be checked by using the button “Data Editor (Browse)”.

- (g) We may ask STATA to generate the residuals of the OLS regression by using the following command:

gen residual = testscr - testscr_hat

Then, a new column called “residual”, which include the OLS residual for each observation, is generated in the data set.

- (h) The modifications to the data set can be stored by using the button “Save”.
- (i) We can use the following command to obtain the scatter plot with the OLS regression line: **scatter testscr str || lfit testscr str**
- (j) Type in the “command” window: **clear** (It clears the memory in STATA, so that we can begin new analysis with other data sets.)

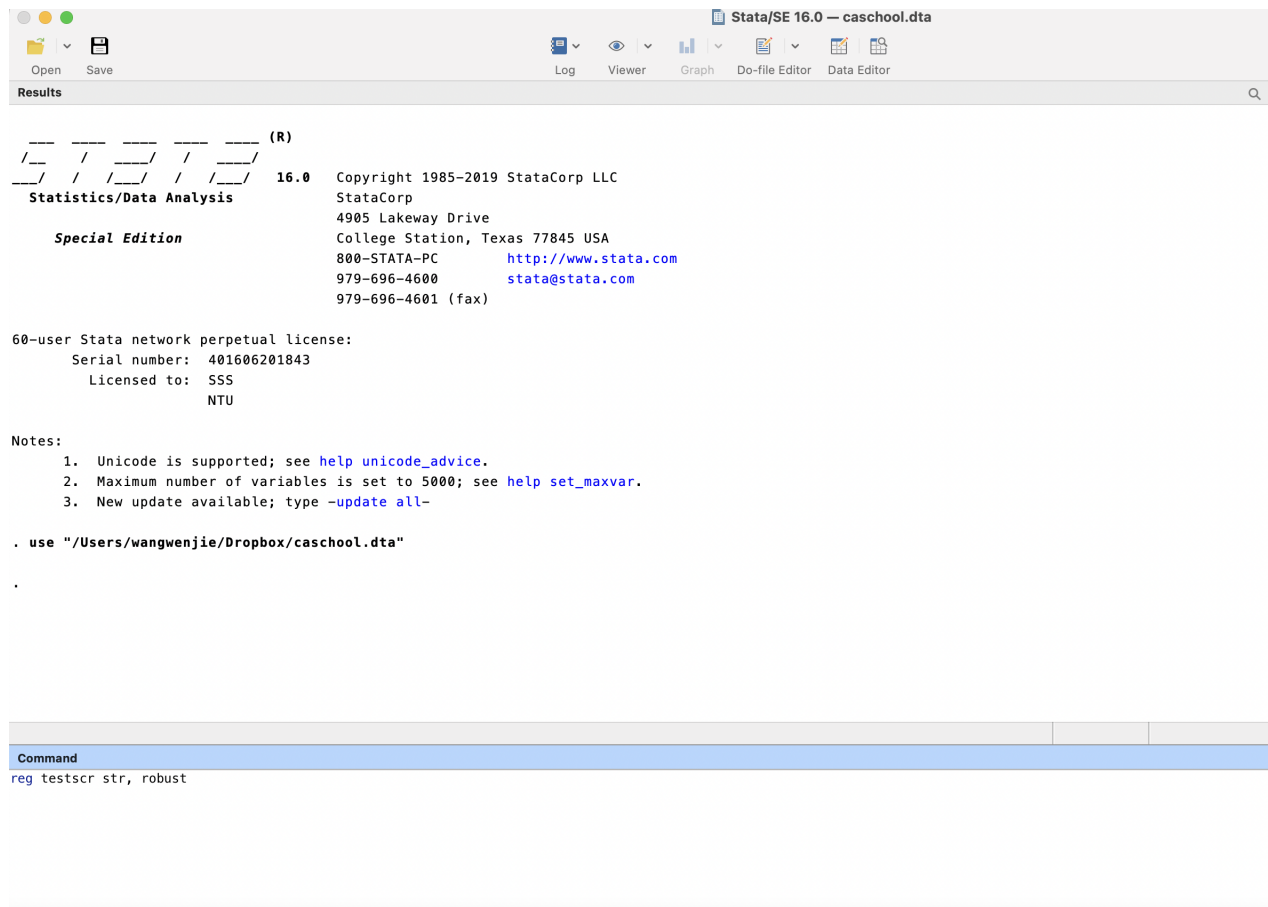


Figure 1

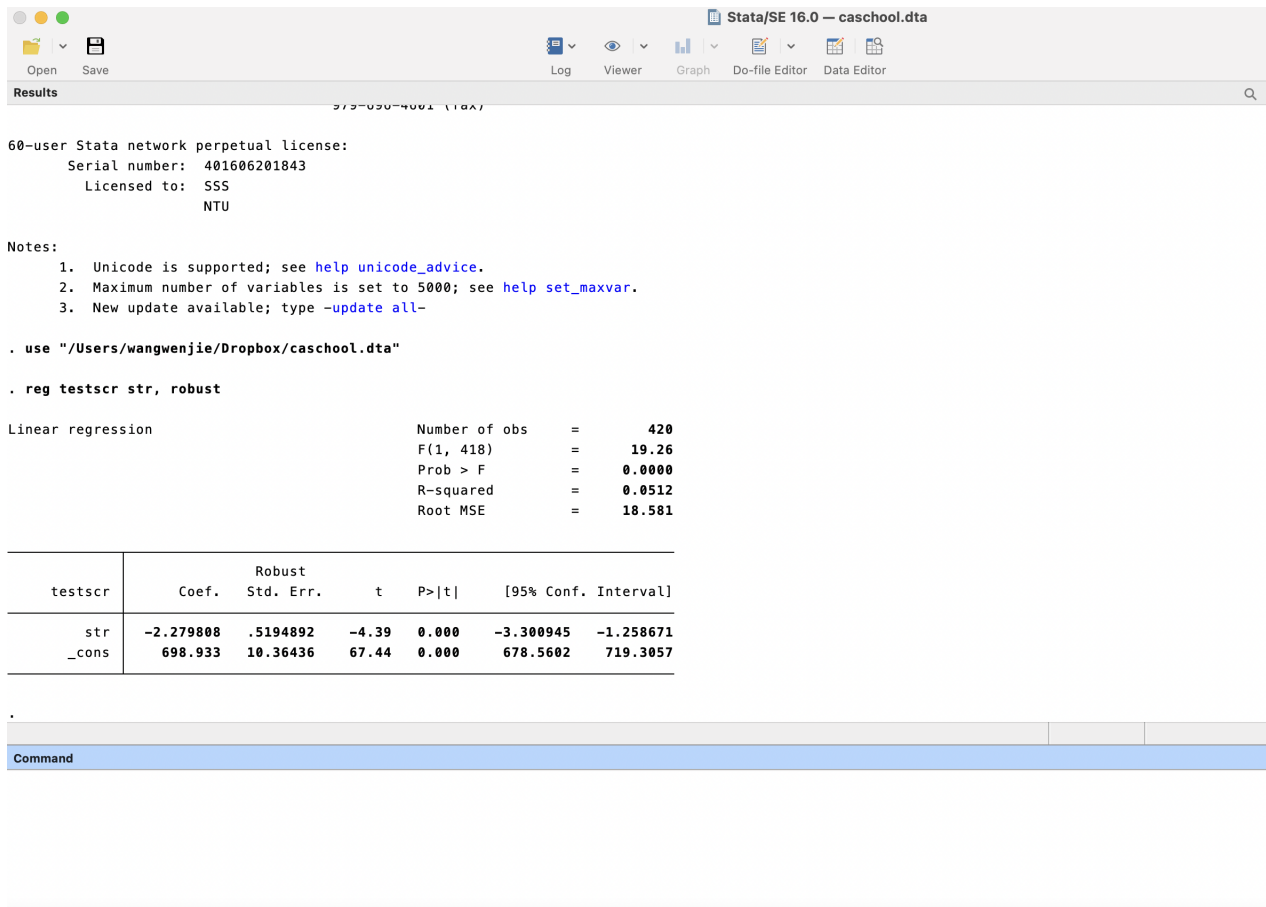


Figure 2